

Student Guide — senior / module-12- movement-of-sun

IKS Curriculum

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Student Workbook

Student Workbook — Movement of the Sun (Senior)

This workbook is yours to write in. Use it for vocab, exercises, and reflection across all 10 days.

Vocabulary tracker

Fill in as we go:

- *Sūrya* — _____
- *Uttarāyaṇa* — _____
- *Dakṣiṇāyana* — _____
- *saṁkrānti* — _____
- *rāśi* — _____
- *sāyana* — _____
- *nirayana* — _____

Day-by-day notes

Day 1 — The Sun's Yearly Path

What I learned:

One question I have:

Day 2 — Axial Tilt and the Seasons

What I learned:

One question I have:

Day 3 — Uttarāyaṇa and Dakṣiṇāyana

What I learned:

One question I have:

Day 4 — Equinoxes — Viṣuva

What I learned:

One question I have:

Day 5 — Twelve Rāśis

What I learned:

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Day 6 — Saṁkrānti Moments

What I learned:

One question I have:

Day 7 — Sāyana vs Nirayana

What I learned:

One question I have:

Day 8 — Precession of the Equinoxes

What I learned:

One question I have:

Day 9 — Cross-Cultural Solar Calendars

What I learned:

One question I have:

Day 10 — Final Share and Assessment

What I learned:

One question I have:

Final reflection

After Day 10, write a paragraph: what is the most important thing you learned in this module?

Day 1 — The Sun’s Yearly Path

Module: Movement of the Sun · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: observe that sunrise direction changes; introduce the ecliptic.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Sūrya* (IAST: Sūrya; lit. “the Sun”)

Primary source (today’s reading)

From *Sūrya-siddhānta* (c. 5th–10th c. CE) — see `sources.md`.

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **The observation.** Project a year-long animation of sunrise direction (or show a static composite of 12 sunrise photos taken from the same spot, one per month). “*The Sun rises in roughly the east — but the EXACT direction shifts north and south through the year.*”
2. **Define the ecliptic.** The Sun’s apparent path against the background stars over one year. A great circle on the celestial sphere, tilted $\sim 23.5^\circ$ from the celestial equator.
3. **Why does it shift?** Earth’s rotational axis is tilted $\sim 23.5^\circ$ relative to its orbital plane. As Earth orbits the Sun, this tilt projects onto our sky as the Sun rising/setting north of due-east in summer (NH), south in winter.
4. **Source.** *Sūrya-siddhānta* ch. 2 — “*sūryasya gatiḥ*” — gives a mean and true solar motion model. Specifies daily and annual displacement on the ecliptic.

5. **Independent task.** Each student looks up their city’s sunrise direction today vs 6 months from today (web). Note the difference in compass bearing.

Activity (15 min)

A scaled version of *Sunrise Direction Tracking* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces The Sun’s Yearly Path.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 2 — Axial Tilt and the Seasons

Module: Movement of the Sun · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: demonstrate with torch+globe why we have summer and winter.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See [lesson-plan.md](#) for any day-specific materials)

Vocabulary introduced today

- *Uttarāyaṇa* (IAST: *Uttarāyaṇa*; lit. “northward solar course (Dec 21 – June 21)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Torch + globe demo.** Hold a globe with the axis tilted $\sim 23.5^\circ$ (mark the North Pole with tape). Shine a strong torch from one direction (the Sun). Slowly rotate the globe and observe which hemispheres get more direct light.
2. **Key observation.** When the Northern Hemisphere tilts TOWARD the Sun (June solstice), it gets more direct sunlight, longer days. Southern Hemisphere experiences the opposite (winter).
3. **Source.** Sūrya-siddhānta and Āryabhaṭīya both note Earth’s globe-shape (*gola*) and the tilt. Āryabhaṭa correctly identifies the rotation cause of day-night ~ 499 CE.
4. **Modern bridge.** NCERT Physics XI ch. 8 gives the modern derivation: Earth’s axial tilt + orbital geometry $\rightarrow$ annual solar declination $\rightarrow$ seasons.
5. **Discussion.** “*Why are January and July temperatures so different — even though Earth is at almost the same distance from the Sun?*” (Answer: tilt, not distance.)

Activity (15 min)

A scaled version of *Sunrise Direction Tracking* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Axial Tilt and the Seasons.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 3 — Uttarāyaṇa and Dakṣiṇāyana

Module: Movement of the Sun · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: identify the dates and direction of each.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Dakṣiṇāyana* (IAST: Dakṣiṇāyana; lit. “southward solar course (June 21 – Dec 21)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Define the two ayanas.**
 - *Uttarāyaṇa* — northward course. Roughly Dec 21 (winter solstice) through Jun 21 (summer solstice). Sun moves NORTH on the sky (Northern Hemisphere).
 - *Dakṣiṇāyana* — southward course. Jun 21 through Dec 21. Sun moves SOUTH.
2. **Cultural importance.** *Uttarāyaṇa* is considered auspicious in many Indian traditions. *Makara Saṅkrānti* (~Jan 14) is celebrated as the start of *Uttarāyaṇa* — but technically the solstice (Dec 21) IS the start. The Jan 14 date reflects the sidereal-zodiac (*nirayana*) convention. We’ll unpack this on Day 7.
3. **Source.** Sūrya-siddhānta defines both ayanas in chapter 2.
4. **Modern bridge.** The Northern Hemisphere’s summer happens during the Sun’s *Uttarāyaṇa* (high northern declination). NCERT Geography IX ch. 3 has the standard diagram.
5. **Identify today.** “*Which ayana are we in today?*” Students compute based on the date. (Northern winter to summer = *Uttarāyaṇa*; rest = *Dakṣiṇāyana*.)

Activity (15 min)

A scaled version of *Sunrise Direction Tracking* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces *Uttarāyaṇa* and *Dakṣiṇāyana*.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See homework.md for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 4 — Equinoxes — Viṣuva

Module: Movement of the Sun · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: identify spring and autumn equinoxes (day = night).

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *saṁkrānti* (IAST: saṁkrānti; lit. “Sun’s crossing from one Rāśi to the next”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Define equinox.** *Viṣuva* in Sanskrit. The two days each year (Mar 21 ± 1 and Sep 22 ± 1) when day length equals night length everywhere on Earth.

2. **Why?** At equinox, Earth's axis is perpendicular to the Sun-Earth line. Sunlight falls equally on both hemispheres. The Sun rises due-east and sets due-west.
3. **Source.** Sūrya-siddhānta names the *viṣuva* points. Varāhamihira (*Bṛhat Samhitā*) discusses them in calendar context. The Vernal Equinox is the traditional anchor of the *sāyana* (tropical) zodiac.
4. **Cross-cultural.** The Roman / Gregorian calendar implicitly aligns to the equinoxes. Easter is computed by lunar-solar reckoning anchored to the Vernal Equinox. Many cultures celebrate the equinoxes — Holi (~Phālguna pūrṇimā, near Vernal Equinox), Mabon (NH autumn).
5. **Observation activity.** Each student notes the sunrise direction this week (or projects a stellarium app). Compare to expected (near-due-east if we're near an equinox).

Activity (15 min)

A scaled version of *Sunrise Direction Tracking* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Equinoxes — Viṣuva.

Wrap-up (5 min)

- One-sentence exit ticket: “*What's one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today's task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 5 — Twelve Rāśis

Module: Movement of the Sun · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: name the 12 rāśis in sanskrit and pair with greek zodiac names.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *rāśi* (IAST: *rāśi*; lit. “1/12th of the ecliptic; zodiacal sign”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Define *rāśi*.** 1/12th of the ecliptic = 30° each. The Sun spends roughly one month in each *rāśi*.
2. **The 12 Sanskrit names.** *Meṣa*, *Vṛṣabha*, *Mithuna*, *Karka*, *Siṃha*, *Kanyā*, *Tulā*, *Vṛścika*, *Dhanus*, *Makara*, *Kumbha*, *Mīna*.
3. **Map to Greek/Western zodiac.** *Meṣa* = Aries, *Vṛṣabha* = Taurus, ... *Mīna* = Pisces. The 12-fold division of the ecliptic is widespread across cultures (Babylonian, Greek, Indian, possibly via cultural contact).
4. **Source.** *Br̥hat Saṃhitā* of Varāhamihira (6th c. CE) systematises the 12 *rāśis*. Earlier references in *Garga Saṃhitā*.
5. **Map current month to current *rāśi*.** Looking at a sidereal *pañcāṅga*, identify which *rāśi* the Sun is in this month. Note the difference from Western (tropical) zodiac dates — we’ll resolve this on Day 7.

Activity (15 min)

A scaled version of *Sunrise Direction Tracking* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Twelve *Rāśis*.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 6 — Saṅkrānti Moments

Module: Movement of the Sun · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: identify makara, karka, and other saṅkrāntis on the calendar.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *sāyana* (IAST: *sāyana*; lit. “tropical zodiac (linked to seasons)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Define saṅkrānti.** The moment the Sun moves from one *rāśi* to the next. ~12 saṅkrāntis per year (one every ~30 days). Two are most culturally prominent: *Makara Saṅkrānti* (Capricorn entry) ~Jan 14, and *Karka Saṅkrānti* (Cancer entry) ~Jul 16.
2. **The Jan 14 / Solstice mystery.** *Makara Saṅkrānti* is celebrated as the start of *Uttarāyaṇa* — but Dec 21 (solstice) is the astronomical start. Why the ~24-day difference? Answer (preview): the celebration uses the *SIDEREAL* (nirayana) zodiac, while the seasons (and solstice) align with the *TROPICAL* (*sāyana*) zodiac. Tomorrow we’ll unpack this.

3. **Source.** Sūrya-siddhānta computes saṅkrāntis as solar longitudes crossing rāśi boundaries.
4. **Cultural saṅkrāntis.** Festivals tied to specific saṅkrāntis: *Makara Saṅkrānti* (Pongal in Tamil Nadu, Lohri in Punjab, Magh Bihu in Assam), *Karka Saṅkrānti* (mid-monsoon).
5. **Class exercise.** Identify the 12 saṅkrānti dates for the current year (use a published pañcāṅga). Plot them on a circular year-wheel.

Activity (15 min)

A scaled version of *Sunrise Direction Tracking* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Saṅkrānti Moments.

Wrap-up (5 min)

- One-sentence exit ticket: “What’s one thing you learned today?”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 7 — Sāyana vs Nirayana

Module: Movement of the Sun · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: distinguish tropical and sidereal zodiacs; introduce ayanāṁśa.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See `lesson-plan.md` for any day-specific materials)

Vocabulary introduced today

- *nirayana* (IAST: nirayana; lit. “sidereal zodiac (linked to fixed stars)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Set up the conflict.** Indian astronomy distinguishes two reference frames:
 - *Sāyana* (with-ayana, “tropical”) — measured from the Vernal Equinox.
 - *Nirayana* (without-ayana, “sidereal”) — measured from a fixed point in the sky (a star, typically near Citrā/Spica).
2. **The drift.** Earth’s axis wobbles (precession of equinoxes), causing the Vernal Equinox point to shift slowly westward against the fixed stars at ~50 arcseconds per year. Over ~2000 years, the cumulative drift is ~28°. *That* is the ~24° gap between Dec 21 (solstice in *sāyana*) and Jan 14 (when the Sun enters Makara in *nirayana*).
3. **Source.** Āryabhaṭīya documents the precession concept; Bhāskara II refines it. Indian *jyotiṣa* preserves the *nirayana* convention; modern astronomy uses *sāyana*.
4. **Why does this matter?** Festival dates anchored to *nirayana* drift slowly relative to the seasons. Over thousands of years, *Makara Saṅkrānti* will shift even further from the December solstice unless re-anchored.
5. **Critical thinking.** “*Should jyotiṣa update its zodiac to sāyana? Or preserve the nirayana convention for cultural continuity?*” Both views have advocates. There’s no single right answer.

Activity (15 min)

A scaled version of *Sunrise Direction Tracking* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces *Sāyana* vs *Nirayana*.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.

- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 8 — Precession of the Equinoxes

Module: Movement of the Sun · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: explain why indian and western zodiac dates differ today.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Sūrya* (IAST: Sūrya; lit. “the Sun”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **The phenomenon.** Earth’s axis wobbles like a slow top, completing one full circle every ~25,800 years. Rate: ~50.3 arcseconds per year. Discovered by Hipparchus ~130 BCE.
2. **Indian computation.** Āryabhaṭīya (499 CE) implicitly includes precession in its longitude calculations. Later Indian astronomers (Bhāskara II, Nīlakaṇṭha) refine the value.
3. **Modern value.** ~50.27”/year — very close to traditional Indian estimates.
4. **Consequence for the zodiac.** The Vernal Equinox point shifts westward through the sidereal zodiac over time. ~2000 years ago, the Vernal Equinox was in Aries (which is why Aries is the “first” zodiac sign in Western astrology). Today, the Vernal Equinox is in Pisces.

5. **Modern source.** NASA / IAU references. Discussion: “*How long until the Sun is in Aquarius at the Vernal Equinox?*” (~600 years.)

Activity (15 min)

A scaled version of *Sunrise Direction Tracking* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Precession of the Equinoxes.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 9 — Cross-Cultural Solar Calendars

Module: Movement of the Sun · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: compare with egyptian, roman, gregorian.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See [lesson-plan.md](#) for any day-specific materials)

Vocabulary introduced today

- *Uttarāyaṇa* (IAST: *Uttarāyaṇa*; lit. “northward solar course (Dec 21 – June 21)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **The Egyptian solar calendar.** 365 days, 12 months of 30 days + 5 epagomenal days. No leap year initially. Synchronised with the Nile flood.
2. **The Roman / Julian / Gregorian.** Julian (45 BCE) — 365.25 days, leap year every 4. Gregorian (1582) — refined to 365.2425, leap years every 4 except centuries unless divisible by 400.
3. **The Indian solar calendar.** *Saura māsa* = time Sun spends in one rāśi. ~30-31 days per month (varies because Sun moves faster near perihelion). Used in Tamil Nadu, Kerala, Bengal — these regions use sidereal solar months even when their festivals are lunar.
4. **The Chinese solar calendar.** 24 *jiéqì* (solar terms) anchored to specific solar longitudes. Spring Equinox, Summer Solstice, etc. are explicit anchor points.
5. **Comparison exercise.** Each student picks 2 traditions and draws a side-by-side year-circle, noting where their respective “new year” falls and what it celebrates.

Activity (15 min)

A scaled version of *Sunrise Direction Tracking* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Cross-Cultural Solar Calendars.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 10 — Final Share and Assessment

Module: Movement of the Sun · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: project share + summative quiz.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Dakṣiṇāyana* (IAST: Dakṣiṇāyana; lit. “southward solar course (June 21 – Dec 21)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Personal solar calendar share (10 min).** Each student shows their year-wheel with 2 solstices, 2 equinoxes, 12 saṅkrāntis, and their own birthday’s rāśi marked.
2. **Vocabulary lightning round (5 min).** Definitions taken from random students: *uttarāyaṇa*, *dakṣiṇāyana*, *viṣuva*, *rāśi*, *saṅkrānti*, *sāyana*, *nirayana*, *ayanāmśa*.
3. **Summative quiz (25 min).** See quizzes/summative.md.
4. **Critical-thinking close.** “*Is the difference between sāyana and nirayana a ‘wrong vs right’ issue or a ‘context-dependent convention’ issue? Which framing is more scientifically honest?*”
5. **Safety reminder.** Never look at the Sun directly. Reaffirm.

Activity (15 min)

A scaled version of *Sunrise Direction Tracking* (see activities/). Today’s slice: focus on the part of the activity that reinforces Final Share and Assessment.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See homework.md for today's task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Writing Prompts

Writing Prompts — Movement of the Sun (Senior)

Use these for in-class writing, homework, or assessment. Each prompt is suitable for a 300-word response.

1. In 3–4 sentences, explain Movement of the Sun to a friend who hasn't taken this module.
2. Choose one Sanskrit term from this module. Define it, then describe one observation from your own life that matches.
3. Compare Movement of the Sun with a similar concept from another module or from your science textbook. What's similar? Different?
4. Identify ONE claim in this module that you'd want to test. How would you test it?
5. What was the most surprising thing in this module? Why did it surprise you?
6. Pick one source cited in this module. Write 2 sentences explaining why it matters.
7. What is one common misconception about Movement of the Sun? Rebut it in 2–3 sentences.
8. Imagine you are teaching this module to a younger student. What would you say first?
9. Read one primary-source passage cited in this module. Translate or paraphrase it; comment on what it adds to your understanding.
10. Identify a contemporary debate about the IKS framework presented here (e.g., is Tirthaji's mathematics genuinely Vedic? are doṣas a valid framework today?). Take a position with two pieces of evidence.

How to use

- Pick **one** prompt per day for the writing-prompts homework.
- Or assign all of them across the module, alternating between in-class and home.
- Senior students may treat any prompt as a 500-word essay topic.

Homework

Homework — Movement of the Sun (Senior)

Light daily tasks. Each takes 10–15 minutes.

1. **Day 1 — The Sun’s Yearly Path** · Write 2 sentences about today’s concept. (~10 min)
2. **Day 2 — Axial Tilt and the Seasons** · Write 2 sentences about today’s concept. (~10 min)
3. **Day 3 — Uttarāyaṇa and Dakṣiṇāyana** · Write 2 sentences about today’s concept. (~10 min)
4. **Day 4 — Equinoxes — Viṣuva** · Write 2 sentences about today’s concept. (~10 min)
5. **Day 5 — Twelve Rāsis** · Write 2 sentences about today’s concept. (~10 min)
6. **Day 6 — Saṁkrānti Moments** · Write 2 sentences about today’s concept. (~10 min)
7. **Day 7 — Sāyana vs Nirayana** · Write 2 sentences about today’s concept. (~10 min)
8. **Day 8 — Precession of the Equinoxes** · Write 2 sentences about today’s concept. (~10 min)
9. **Day 9 — Cross-Cultural Solar Calendars** · Write 2 sentences about today’s concept. (~10 min)
10. **Day 10 — Final Share and Assessment** · Write 2 sentences about today’s concept. (~10 min)

Notes

- Homework is intended to consolidate what was learned in class, not introduce new content.
- If a child struggles, the parent should consult the teacher rather than push.
- Capstone project days (Module 15) may have lighter daily homework so students can focus on the project.